Polarisation of light

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In this experiment, several polarization phenomena of light were investigated. Malus's law was verified by measuring the transmitted intensity as a function of the analyzer angle, showing the expected $\cos^2(\theta)$ dependence. The birefringence of calcite was observed through the formation of ordinary and extraordinary rays. Using a $\lambda/4$-plate, linear polarization was transformed into circular and elliptical polarization states and described with the Jones formalism. The optical activity of sucrose solutions was analyzed with a half-shadow polarimeter, yielding an unknown concentration of $c = (64.3 \pm 6.8)\,\mathrm{g\,l^{-1}}$. Furthermore, the Faraday effect in glass was examined for different wavelength ranges. The measured Verdet constants decreased from $(23.37 \pm 0.40)\,\mathrm{rad\,T^{-1}m^{-1}}$ at 410 nm to $(13.62 \pm 0.24)\,\mathrm{rad\,T^{-1}m^{-1}}$ at 595 nm, in agreement with theoretical expectations.

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1 Introduction

Light is an electromagnetic wave whose electric field can exhibit a specific orientation, a property known as polarization. The polarization of light is a fundamental aspect of optics and plays an important role in a wide range of scientific and technological applications. Examples include polarized sunglasses that reduce glare, optical microscopy techniques that enhance image contrast, modern telecommunications systems that use polarization for signal transmission, and emerging quantum technologies where polarization states can serve as carriers of information.
The aim of this experiment is to investigate the fundamental properties of polarized light. Different methods for generating polarized light will be explored, and the effects of various optical elements on its polarization state will be examined. Furthermore, techniques for measuring and analyzing polarization will be introduced. Through these investigations, a deeper understanding of the behavior of polarized light and its practical applications in optical systems will be developed.

2 Malus's law

2.1 Theory of Malus's law

Malus's law describes the intensity of polarized light after passing through a polarizer, it's derivation can be found in the appendix 9. It states that the intensity I of the transmitted light is related to the intensity I_0 of the incident light and the angle θ between the light's initial polarization direction and the axis of the polarizer by the equation:

$$I = I_0 \cos^2(\theta) \tag{1}$$

After passing through the polarizer, the light is polarized along the axis of the polarizer. The wave components that are not transmitted are absorbed by the polarizer, resulting in a decrease in intensity.

2.2 Experimental setup for Malus's law

The unpolarized light of an incandescent lamp was first vertically polarized using a polarizer. The polarized light then passed through a second polarizer, called the analyzer, which can be rotated to change the angle θ between the polarization axis of the polarizer and the analyzer. The angle θ was varied from 0° to 180° in increments of 10°, and the corresponding intensity values were measured using a photocell.

2.3 Results of Malus's law

As one can see in fig. 1, for low intensity values, the measured data points fit better to fitted curve, whereas for the angle from $\theta = 0° - 90°$ the data points fit good tho the theoretical curve. This indicates, that the measured intensity values are described sufficiently enough by Malus's law, but there are some systematic errors in the measurement, which can be caused by the fact that the polarizers are not ideal and do not completely absorb the light that is not transmitted. Another source of error could be, that the room was not completely dark, which could

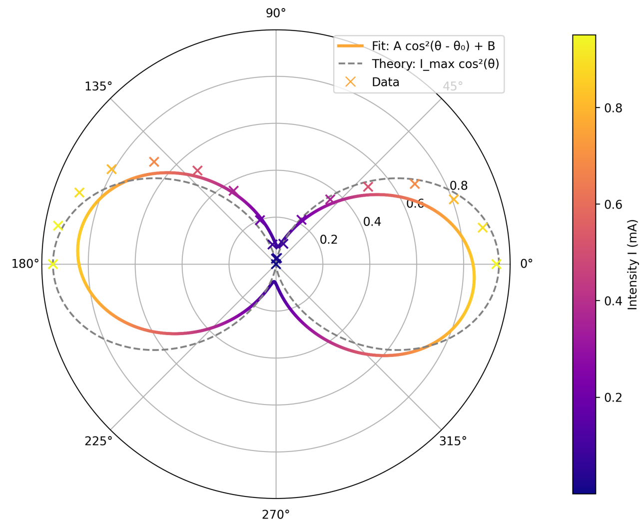

Figure 1: Measured intensity values and the fitted curve with fit $I = A\cos^2(\theta - \theta_0) + B$, $A = 0.7720mA$, $\theta_0 = -5.95°$, $B = 0.0761mA$.

have led to additional light being measured by the photocell. Additionally, the angle θ was not measured with high precision, which could have also contributed to the deviations from the theoretical curve.

3 Birefringence of a calcite crystal

3.1 Theory of birefringence

Birefringence, also known as double refraction, is a phenomenon that occurs in certain anisotropic materials, such as calcite crystals. When light enters a birefringent material, it splits into two rays: the ordinary ray and the extraordinary ray. These rays experience different refractive indices due to the anisotropic nature of the material, resulting in different propagation speeds and directions. The ordinary ray follows Snell's law, while the extraordinary ray does not, leading to a separation of the two rays within the crystal.

3.2 Experimental setup for birefringence

In this experiment, a calcite crystal was placed in the path of an unpolarized light beam. As the light passed through the calcite crystal, it split into the ordinary and extraordinary rays. A analyzer was placed behind the crystal to observe the polarization states of the two rays. The rays were projected onto a screen. By rotating the polarizer and observing the changes in intensity and direction of the rays, we can analyze the birefringent properties of the calcite crystal.

3.3 Results of birefringence

4 $\lambda/4$-Plate

4.1 Theory of a $\lambda/4$-plate

A $\lambda/4$-plate, also known as a quarter-wave plate, is an optical device that introduces a phase shift of $\pi/2$ (90 degrees) between the orthogonal components of polarized light.

This phase shift arises from birefringence: the plate is made of an anisotropic crystal, in which light polarized parallel to the optic axis (extraordinary ray) experiences a different refractive index n_e than light polarized perpendicular to the optic axis (the ordinary ray), which experiences n_o. Since the two polarization components travel through the same physical thickness d but at different phase velocities $v_o = c/n_o$ and $v_e = c/n_e$, they accumulate different optical path lengths, and hence a relative phase difference, when passing through the crystal.

For a plane wave of vacuum wavelength λ traveling a distance d through the crystal, each component picks up a phase

$$\phi_o = \frac{2\pi}{\lambda} n_o d, \qquad \phi_e = \frac{2\pi}{\lambda} n_e d. \tag{2}$$

The phase difference between the two components after traversing the plate is therefore

$$\delta = \phi_e - \phi_o = \frac{2\pi}{\lambda}(n_e - n_o)\,d = \frac{2\pi}{\lambda}\Delta n\,d, \tag{3}$$

where $\Delta n = n_e - n_o$ is the birefringence of the material.
For a quarter-wave plate, we require $\delta = \pi/2$. Setting this equal to the expression above and solving for d:

$$\frac{2\pi}{\lambda}\Delta n\,d = \frac{\pi}{2} \quad \Longrightarrow \quad d = \frac{\lambda}{4\,|\Delta n|}. \tag{4}$$

This is the minimum thickness. In practice, plates are often made thicker for mechanical robustness which still produces a phase difference of $\pi/2$ modulo 2π, since adding any integer number of full wavelengths of relative delay does not change the resulting polarization state.

4.2 Experimental setup for a $\lambda/4$-plate

In this experiment a $\lambda/4$-plate made of quartz was used. The light of an incandescent lamp was first linearly polarized using a polarizer. The linearly polarized light then passed through the $\lambda/4$-plate which was oriented at the angles $\alpha = 0°, -45°, -90°$ with respect to the polarization direction of the incident light. After passing through the $\lambda/4$-plate, the light was analyzed using a analyzer which was rotated to observe the changes in intensity and polarization state of the transmitted light.

Afterwards the $\lambda/4$-plate was oriented at $\alpha = -45°$ and another identical $\lambda/4$-plate was placed in the ray path, such that its optical axis was oriented parallel to the first $\lambda/4$-plate.

4.3 Results of a $\lambda/4$-plate

For $\alpha = 0°$ and $\alpha = -90°$, the transmitted light remained linearly polarized because the light is polarized along the same axis to one of the oplical axes and excites only that component. Thus no relative phase shift occurs and the polarization state does not change. For $\alpha = -45°$, the transmitted light became circularly polarized because the light is polarized at 45 degrees to the optical axes of the $\lambda/4$-plate, exciting both components equally. The $\lambda/4$-plate introduces a phase shift of $\pi/2$ between the two components, resulting in circular polarization. At any other angle $\alpha =]0, -90°[\ \setminus -45°$ the transmitted light became elliptically polarized because the light is polarized at an angle that is not aligned with the optical axes of the $\lambda/4$-plate, exciting both components but with different amplitudes.
When the second $\lambda/4$-plate was added with its optical axis parallel to the first one, tthe system is equivalent to a $\lambda/2$-plate, which introduces a phase shift of π between the two components. The rotation angle is twice the angle between the incident polarization and the optical axis of the plate. The light remains linearly polarized but with a rotated polarization plane. If the quartz $\lambda/4$-plate is replaced by a plate made of calcite, the thickness of the plate would need to be adjusted to achieve the same phase shift of $\pi/2$ because the birefringence of calcite is different from that of quartz.

5 Jones-Formalism

The Jones formalism can be used to describe mathematically the polarization state of light and the effect of polarizers and wave plates. Here the polarization state is represented as a two-dimensional vector and the optical elements are represented as a 2x2 matrix.

Linearly polarized light in x-direction:	$\vec{J}_x =$	$\begin{pmatrix}1\\0\end{pmatrix}$
Linearly polarized light in y-direction:	$\vec{J}_y =$	$\begin{pmatrix}0\\1\end{pmatrix}$
Linearly polarized light at 45°:	$\vec{J}_{45} =$	$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$
Right circularly polarized light:	$\vec{J}_R =$	$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-i\end{pmatrix}$
Left circularly polarized light:	$\vec{J}_L =$	$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\i\end{pmatrix}$

The Jones matrix for the $\lambda/4$-plate with the fast axis along the x-direction is given by:

$$\mathbf{M}_{\lambda/4} = \frac{1}{\sqrt{2}}\begin{pmatrix}1-i & 0\\ 0 & 1+i\end{pmatrix}$$

This matrix introduces a phase shift of $\pi/2$ between the x and y components of the electric field, which is characteristic of a quarter-wave plate. When linearly polarized light at 45 degrees passes through this $\lambda/4$-plate, it becomes circularly polarized. Conversely, circularly polarized light can be converted back into linearly polarized light by passing it through the same $\lambda/4$-plate swith the appropriate orientation.
Applying $M_{\lambda/4}$ to light linearly polarized at 45°:

$$\begin{aligned} M_{\lambda/4}\,\vec{J}_{45} &= \frac{1}{\sqrt{2}}\begin{pmatrix}1-i & 0\\ 0 & 1+i\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix}1-i\\1+i\end{pmatrix}. \end{aligned} \tag{5}$$

Using $1-i = \sqrt{2}\,e^{-i\pi/4}$ and $1+i = \sqrt{2}\,e^{i\pi/4}$, this becomes

$$\begin{aligned} \frac{1}{2}\begin{pmatrix}\sqrt{2}\,e^{-i\pi/4}\\ \sqrt{2}\,e^{i\pi/4}\end{pmatrix} &= e^{-i\pi/4}\,\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ e^{i\pi/2}\end{pmatrix} \\ &= e^{-i\pi/4}\,\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix} \\ &= e^{-i\pi/4}\,\vec{J}_L. \end{aligned} \tag{6}$$

Since an overall phase factor $e^{-i\pi/4}$ has no physical effect on the polarization state, this shows that

$$M_{\lambda/4}\,\vec{J}_{45} \;\propto\; \vec{J}_L, \tag{7}$$

i.e. linearly polarized light at 45° is converted into left circularly polarized light, as claimed.

Conversely, applying $M_{\lambda/4}$ to left circularly polarized light:

$$\begin{aligned} M_{\lambda/4}\,\vec{J}_L &= \frac{1}{\sqrt{2}}\begin{pmatrix}1-i & 0\\ 0 & 1+i\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1-i\\ i(1+i)\end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix}1-i\\ -1+i\end{pmatrix} = \frac{1-i}{2}\begin{pmatrix}1\\ -1\end{pmatrix}. \end{aligned} \tag{8}$$

Writing $1-i = \sqrt{2}\,e^{-i\pi/4}$, this is

$$M_{\lambda/4}\,\vec{J}_L = e^{-i\pi/4}\,\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -1\end{pmatrix} \;\propto\; \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -1\end{pmatrix}, \tag{9}$$

which (up to the irrelevant global phase) is linearly polarized light at $-45°$. This confirms that circularly polarized light is converted back into linearly polarized light by the same $\lambda/4$-plate.

6 Circular birefringence of sucrose

In many fields of study concentration analysis of unknown solutions is fundamental. To quickly

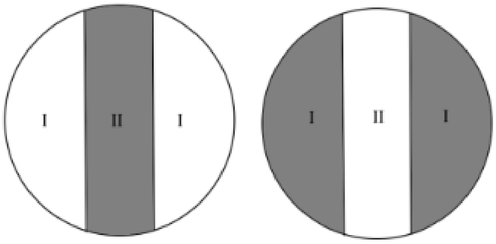

Figure 2: Looking through the polarimeter, there are two fields visible, one at the edges and one in the middle. Field I has a slightly different polarisation angle than field II. The light from both fields passes through an adjustable polarisation filter, causing a difference of brightness.

determine the concentration of a solution, one can use the method of half-shadow polarimeter. This method is based on the fact that certain substances, such as sucrose, exhibit circular birefringence, which causes the plane of polarization of light to rotate as it passes through the solution. The angle of rotation is directly proportional to the concentration of the solution and the path length of the light through it. By measuring the angle of rotation, one can determine the concentration of the solution.

The key component that gives the half-shadow polarimeter its name is a small auxiliary plate placed directly behind the first polarizer , covering only half of the beam's cross-section. This plate is typically either a thin quartz plate or simply a second polarizing film whose transmission axis is rotated by a small angle 2α relative to the transmission axis of the main polarizer, where α is the so-called half-shade angle. As a result, the light entering the sample cell is split into two halves with slightly different polarization directions:

- Field I (e.g. the outer half of the field of view) is polarized along the axis of the main polarizer, at angle 0.
- Field II (e.g. the central half, behind the auxiliary plate) is polarized at an angle 2α relative to Field I.

Both halves of the beam then pass through the sample and through the analyzer before reaching the eye. Because the two halves of the field have different polarization directions, rotating the analyzer changes the brightness of the two halves differently: for most analyzer orientations, one half appears brighter than the other. Only when the analyzer's transmission axis lies exactly along the bisector of the two polarization directions do both halves appear equally dark, since the analyzer is then tilted by the same angle α relative to each half. This equal-darkness condition can be located far more precisely by the eye than the single minimum of brightness used in a simple polarimeter, since the human eye is much better at comparing the relative brightness of two adjacent fields than at judging absolute minimum brightness. This is the central advantage of the half-shade design.

6.1 Measurements and results

When an optically active sample is placed in the beam path, it rotates the polarization of both halves of the field by the same angle φ, since both halves traverse the same path length and concentration of the solution. The angle bisector between the two field polarizations is therefore also rotated by φ, and the analyzer must be rotated by the same angle φ to restore the equal-darkness condition. Measuring this required rotation of the analyzer thus directly yields the optical rotation angle φ introduced by the sample.

For that method, the penumbral angle α of the used polarimeter has to be determined, by measuring the position of minimal brightness in Field I and Field II individually and then subtracting the two corresponding analyzer angles from each other. Half of this difference corresponds to the half-shade angle α, and the bisector of the two angles defines the zero point used as the reference for subsequent concentration measurements.

The penumbral angle of the used polarimeter was determined to be $\psi = (11.60 \pm 0.73)^\circ$.
Now the concentration of an unknown sucrose solution had to be determined. For that, we sent light with wavelength $\lambda = 595nm$ through a solution of sucrose in water with concentration $c_{100} = 100g/l$, examined it's optical rotation which resulted $\theta_{100} = (14.00 \pm 0.75)^\circ$ and afterwards determined the rotation of a solution with unknown concentration to $\theta_{unknown} = (9.00 \pm 0.59)^\circ$.
The sucrose concentration of the unknown solution can be calulated using the formula:

$$c_{unknown} = \frac{\theta_{unknown}}{\theta_{100}/c_{100}} \tag{10}$$

With the measured values, the concentration of the sucrose was determined to $c_{unknown} = (64.3 \pm 6.8)\frac{g}{l}$. This seems reasonable for the amount of water, the c_{100} solution was diluted with.

7 Measuring the Faraday effect inside a glass body

In the Faraday effect, the plane of polarisation of linearly polarised light is rotated while propagating through a material placed inside a magnetic field. For an approximately constant magnetic field, the following relation holds:

$$\Theta = V \cdot B_{\mathrm{eff}} \cdot l$$

where Θ is the measured rotation angle, V the Verdet constant, B_{eff} the effective magnetic flux density, and l the length of the glass sample.
In this experiment, a glass body with a length of $l = (30.0 \pm 0.5)\,\mathrm{mm}$ was investigated.

7.1 Determination of the Effective Magnetic Field

First, the magnetic field along the optical axis was measured using a Hall probe. Since the magnetic field inside the solenoid is not perfectly homogeneous, an effective magnetic field was determined from the measured field profile.
The effective magnetic field is defined as:

$$B_{\mathrm{eff}} = \frac{1}{l} \int_0^l B(x)\, dx$$

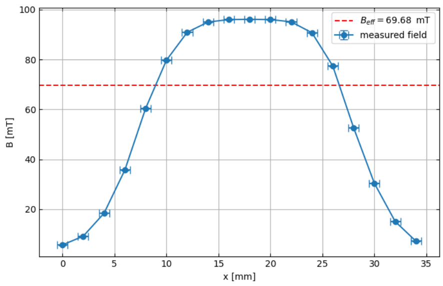

Figure 3: Magnetic flux density $B(x)$ as a function of the position x along the optical axis. Raw data shown in tab. 3

From this, the effective magnetic field is determined to be:

$$B_{\mathrm{eff}} = (69.7 \pm 1.1)\,\mathrm{mT}$$

based on the maximum magnetic field ,$B_{\mathrm{max}} = (96.1 \pm 0.05)\,\mathrm{mT}$, the following value is obtained for the parameter α:

$$\alpha = \frac{B_{\mathrm{eff}}}{B_{\mathrm{max}}} = 0.725 \pm 0.012$$

This factor is later used to determine the effective magnetic field for other coil currents from the measured maximum magnetic field. Therefore, we assume that α remains constant for different currents, as the magnetic field geometry should not should not significantly depend on the coil current.

7.2 Magnetic flux density per current

Afterwards, the effective magnetic flux density was determined for different currents. Plotting B_{eff} as a function of the current I reveals an approximately linear dependence.

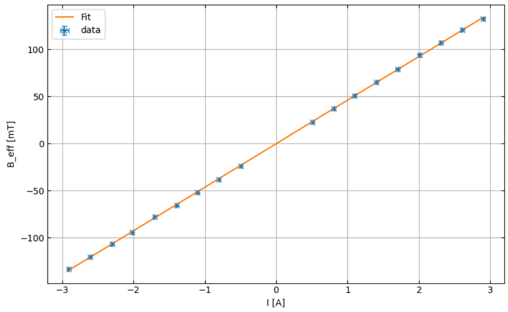

Figure 4: Maximal flux densities for different currents. The proportionality constant between the strongest magnetic flux density and the solenoid current is determined by conducting a linear regression.

As shown in Fig. 4, B_{max} exhibits an approximately linear dependence on I. Therefore, a linear function was fitted, yielding the slope to be $C_B = \frac{dB_{\mathrm{max}}}{dI} = (46,245 \pm 0.013)\,\mathrm{mT/A}$.

7.3 Faraday Rotation per current

For different wavelength ranges, the rotation angle Θ of the plane of polarisation was measured as a function of the effective magnetic field B_{eff}.
The analyser was rotated until the signal of the photodiode reached a minimum. The corresponding angle was interpreted as the rotation angle caused by the Faraday effect.
For each wavelength range, a linear function fit was performed.

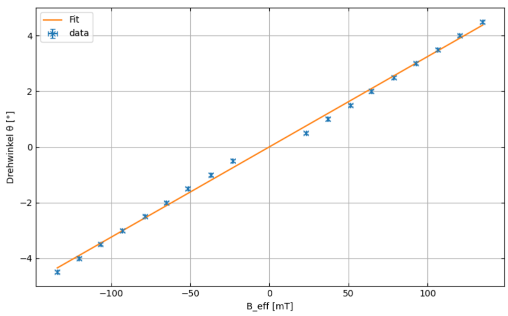

Figure 5: Rotation angle Θ as a function of the effective magnetic field B_{eff} for the wavelength range 400–560 nm. The slope of the linear fit is used to calculate the Verdet constant.

Filter range	slope / $^\circ\,\mathrm{mT}^{-1}$
360–460	(0.04017 ± 0.00014)
400–560	(0.03246 ± 0.00014)
560–630	(0.02341 ± 0.00014)

Table 1: Slopes of the linear fits for the different wavelength ranges.

7.4 Calculation of the Verdet Constant

Using the relation

$$\Theta = V B_{\text{eff}} l$$

the Verdet constant can be calculated from the slope of the linear fits:

$$V = \frac{\Theta}{B_{\text{eff}} l}$$

Since the slope m is obtained in units of $\frac{\circ}{\mathrm{mT}}$, a conversion into SI units is necessary:

$$V = m \cdot \frac{\pi}{180} \cdot 1000 \cdot \frac{1}{l}$$

with

$$l = 0.030\,\mathrm{m}$$

The calculated Verdet constants are summarised in Table 2.

Filter range	λ / nm	V / rad(Tm)$^{-1}$
360–460	410 ± 50	(23.37 ± 0.40)
400–560	480 ± 80	(18.89 ± 0.32)
560–630	595 ± 35	(13.62 ± 0.24)

Table 2: Verdet constants for different wavelength ranges.

7.5 Dispersion of the Verdet Constant

Finally, the Verdet constant was plotted as a function of the mean wavelength of the corresponding interference filter.

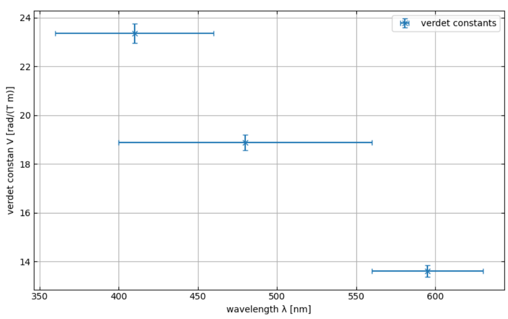

Figure 6: Dispersion of the Verdet constant as a function of the wavelength.

The measurements show that the Verdet constant decreases with increasing wavelength. This behaviour is expected for dispersive optical media such as N-SF6 glass. Shorter wavelengths experience a stronger rotation of the plane of polarisation than longer wavelengths.

The Faraday effect can be understood as a magnetically induced circular birefringence: in the presence of an external magnetic field, the medium acquires slightly different refractive indices n_+ and n_- for left- and right-circularly polarized light, and this difference is what produces the rotation of the plane of polarization. The Verdet constant is therefore closely tied to how strongly, and how quickly, the refractive index of the medium changes with wavelength – i.e. to the material's dispersion $dn/d\lambda$. This relationship is expressed approximately by the Becquerel formula,

$$V(\lambda) \;\approx\; -\frac{e}{2m_e c}\,\lambda\,\frac{dn}{d\lambda}, \tag{11}$$

which shows that the Verdet constant is proportional to the slope of the refractive index curve, $dn/d\lambda$, rather than to the refractive index itself.

In the region of normal dispersion, the refractive index $n(\lambda)$ decreases with increasing wavelength, and it does so more steeply at shorter wavelengths than at longer ones.

Physically, this means that at shorter wavelengths, light interacts more strongly with the bound electrons of the glass (closer to their resonance/absorption frequencies in the UV), leading to a larger magnetically induced splitting between n_+ and n_-, and hence a larger rotation angle per unit field and path length.

The data point corresponding to the wavelength range 360–460 nm should be interpreted carefully, since several measured values deviate more strongly from the linear behaviour. Possible reasons include reduced light intensity in this wavelength range, alignment uncertainties, or thermal effects caused by heating of the coils.

The Faraday effect was successfully observed for all investigated wavelength ranges. In each case, an approximately linear dependence between the rotation angle and the effective magnetic field was found.

The calculated Verdet constants satisfy the qualitative relation

$$V_{360-460} > V_{400-560} > V_{560-630}$$

which means that the Verdet constant decreases with increasing wavelength, as expected from theory.

8 Conclusion

In this experiment, several important phenomena related to the polarization of light were successfully investigated. Malus's law was experimentally verified,

showing the expected cosine-squared dependence between transmitted intensity and the angle between polarizer and analyzer. Although small deviations from the theoretical prediction occurred, mainly due to non-ideal optical components and experimental uncertainties, the overall agreement confirmed the validity of the model.
The birefringence of calcite crystals demonstrated how anisotropic materials split light into ordinary and extraordinary rays with different polarization states. Furthermore, the behavior of quarter-wave plates illustrated the conversion between linear, circular, and elliptical polarization, depending on the orientation of the optical axis.
Using the Jones formalism, polarization states and optical elements could be described mathematically, providing a useful theoretical framework for understanding the observed effects. The experiment with sucrose solutions showed that circular birefringence can be applied to determine concentrations through optical rotation measurements.
Finally, the Faraday effect in glass was successfully observed. A linear relationship between the rotation angle and the magnetic field strength was found for all investigated wavelength ranges. The calculated Verdet constants decreased with increasing wavelength, which agrees with the theoretical expectations for dispersive optical media.
Overall, the experiment provided a comprehensive understanding of polarization phenomena and demonstrated the close connection between theoretical optics and experimental observations.

9 Appendix

Derivation of Malus's Law

Consider linearly polarized light with intensity I_0 and polarization direction $\vec{e}_P$ incident on an analyzer whose transmission axis $\vec{e}_A$ is rotated by an angle θ relative to $\vec{e}_P$. Only the component of the electric field along $\vec{e}_A$ is transmitted.
The incident electric field can be written as

$$\vec{E} = E_0\,\vec{e}_P, \tag{12}$$

where E_0 is the amplitude. The transmitted component along $\vec{e}_A$ is

$$E_t = \vec{E}\cdot\vec{e}_A = E_0\cos\theta. \tag{13}$$

Since the intensity is proportional to the square of the field amplitude, $I \propto E^2$, it follows that

$$I(\theta) = I_0\cos^2\theta \tag{14}$$

Mean Intensity after the Analyzer

If unpolarized light of intensity I_{in} is incident on the polarizer, an ideal polarizer transmits exactly half of it:

$$I_0 = \frac{I_{\text{in}}}{2}. \tag{15}$$

This follows from averaging $\cos^2\theta$ over all polarization directions:

$$\langle\cos^2\theta\rangle = \frac{1}{2\pi}\int_0^{2\pi}\cos^2\theta\,\mathrm{d}\theta = \frac{1}{2\pi}\int_0^{2\pi}\frac{1+\cos 2\theta}{2}\,\mathrm{d}\theta = \frac{1}{2}. \tag{16}$$

Therefore, the intensity of the light incident on the analyzer is

$$I_0 = \frac{I_{\text{in}}}{2} \tag{17}$$

i.e. the intensity after the polarizer is exactly half of the incoming intensity, independent of the orientation of the analyzer.

Data Faradays effect

x /mm	$B(x)$ /mT
0	5.7
2	9.2
4	18.4
6	35.7
8	60.4
10	79.7
12	90.9
14	95
16	96
18	96.1
20	96
22	95.1
24	90.7
26	77.4
28	52.7
30	30.4
32	15.1
34	7.2

Table 3: Magnetic field $B(x)$ as a function of position x.

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